A DOMAIN WITH A NONZERO IDEAL PRINCIPAL AT EVERY PRIME BUT NOT INVERTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The terminology paragraph on 'locally principal' (2560-2571) is the author's terminological remark and is not needed for the argument; the counted chain uses only the two spans.

Editorial note. The label lemma-locally-principal-not-invertible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2536-2541 (section-nonfree): R = Q[x] and M = sum over n of (1/(x-n)) R as a submodule of the fraction field; M is not finitely generated, but for every prime p of R we have M_p = R_p as an R_p-module (the author's stalk computation stated in the section)
- Context span 2574-2597: A = Sym*_R(M) (an example of a non-Noetherian domain with Noetherian local rings, author's footnote) and I = MA = the direct sum of the Sym^d_R(M) for d >= 1; M not finitely generated implies I is not finitely generated as an ideal, and an invertible module is finitely generated, so I is not invertible; for every prime p of R, M_p = R_p gives A_p = Sym*_{R_p}(R_p) = R_p[T] as a graded R_p-algebra, so I_p is generated by the nonzerodivisor T; hence for every prime q of A the ideal I_q is generated by a single element

Source reference. Original section 065J: sources/stacks-examples/pinned-original-source.tex, source lines 2533–2533. Prerequisite pointer invoked at 2623; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let $R = \mathbf{Q}[x]$. Set $M = \sum_{n \in \mathbf{N}} \frac{1}{x-n} R$ as a submodule of the fraction field of R. Then M is not finitely generated, but for every prime $\mathfrak{p}$ of R we have $M_\mathfrak{p} \cong R_\mathfrak{p}$ as an $R_\mathfrak{p}$-module.

An example of a similar flavor is $R = \mathbf{Z}$ and $M = \sum_{p \text{ prime}} \frac{1}{p}\mathbf{Z} \subset \mathbf{Q}$, which equals the set of fractions $\frac{a}{b}$ with b nonzero and squarefree.

Let $R = \mathbf{Q}[x]$ and let $M = \sum \frac{1}{x-n} R$ be the module constructed in Section 065J. Consider the ring[1]

$$A = \mathrm{Sym}^*_R(M)$$

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[1]The ring A is an example of a non-Noetherian domain whose local rings are Noetherian.

and the ideal $I = MA = \bigoplus_{d \geq 1} \mathrm{Sym}^d_R(M)$. Since M is not finitely generated as an R-module we see that I cannot be generated by finitely many elements as an ideal in A. Since an invertible module is finitely generated, this means that I is not invertible. On the other hand, let $\mathfrak{p} \subset R$ be a prime ideal. By construction $M_\mathfrak{p} \cong R_\mathfrak{p}$. Hence

$$A_\mathfrak{p} = \mathrm{Sym}^*_{R_\mathfrak{p}}(M_\mathfrak{p}) \cong \mathrm{Sym}^*_{R_\mathfrak{p}}(R_\mathfrak{p}) = R_\mathfrak{p}[T]$$

as a graded $R_\mathfrak{p}$-algebra. It follows that $I_\mathfrak{p} \subset A_\mathfrak{p}$ is generated by the nonzerodivisor T. Thus certainly for any prime ideal $\mathfrak{q} \subset A$ we see that $I_\mathfrak{q}$ is generated by a single element.

Lemma 0.1. *There exists a domain A and a nonzero ideal $I \subset A$ such that $I_\mathfrak{q} \subset A_\mathfrak{q}$ is a principal ideal for all primes $\mathfrak{q} \subset A$ but I is not an invertible A-module.*

Proof. See discussion above. □